



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING – *Written Exam* - 11/01/2018

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = As[n] + B(-1)^n s[n] + W[n], \quad n = 0, 1, \dots, N-1,$$

where A and B are the amplitudes of the Signals of Interest (SOI) and they are modelled as unknown deterministic parameters, the sample size N is an even number, $s[n]$ is the signal shape, perfectly known, we assume here that it is a constant signal, i.e. $s[n] = 1$, for $0 \leq n \leq N-1$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_W(e^{j2\pi f}) = \sigma_W^2$, where σ_W^2 is a-priori known.

- (1) Derive the joint probability density function (pdf) $f_{\mathbf{X}}(\mathbf{x}; A, B)$ of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$ and the log-likelihood function $\ln L(A, B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B)$.
- (2) Under the assumption that N is even, derive the joint Maximum Likelihood (ML) estimator of the parameters A and B , the pdf of the estimation errors $\varepsilon_A \triangleq A - \hat{A}_{ML}$ and $\varepsilon_B \triangleq B - \hat{B}_{ML}$, and the Mean Square Error (MSE) of the estimators, $MSE_A \triangleq E\{\varepsilon_A^2\}$ and $MSE_B \triangleq E\{\varepsilon_B^2\}$. Establish whether the two estimators are **biased** and **consistent**.
- (3) Derive the Cramér-Rao lower bound (CRB) on the estimate of A and B , compare them with the MSEs of the ML estimators and establish whether the two ML estimators are **efficient**.
- (4) Explain why the MSEs on the estimate of A and B are the same. Calculate which is the minimum value of N that guarantees that the standard deviation of ε_B is lower than the 10% of the true value of B , under the assumption that $B=5$ and $\sigma_W^2=10$.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

Solution

(1) $\mathbf{X}$ is a Gaussian random vector (deterministic signal in AWGN). The data can be modeled according to the deterministic linear model in AWGN:

$$\mathbf{X} = A \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + B \begin{bmatrix} (-1)^0 \\ (-1)^1 \\ \vdots \\ (-1)^{N-1} \end{bmatrix} + \mathbf{W} = A \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + B \begin{bmatrix} 1 \\ -1 \\ \vdots \\ -1 \end{bmatrix} + \mathbf{W} = A\mathbf{1} + B\mathbf{p} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$\boldsymbol{\theta}_{2 \times 1} \triangleq \begin{bmatrix} A \\ B \end{bmatrix}, \quad \mathbf{H}_{N \times 2} \triangleq [\mathbf{1} \quad \mathbf{p}]$$

$$\mathbf{X} \in N(\mathbf{A}\mathbf{1} + B\mathbf{p}, \sigma_w^2 \mathbf{I})$$

$$\ln L(A, B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B) = k - \frac{1}{2\sigma_w^2} (\mathbf{x} - (\mathbf{A}\mathbf{1} + B\mathbf{p}))^T (\mathbf{x} - (\mathbf{A}\mathbf{1} + B\mathbf{p})) = k - \frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})$$

(2) For the deterministic linear model in AWGN, we easily get:

$$\hat{\boldsymbol{\theta}}_{ML} = \mathbf{H}^{\#} \mathbf{X} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X}$$

$$\mathbf{H}^{\#} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T = \left(\begin{bmatrix} \mathbf{1}^T \\ \mathbf{p}^T \end{bmatrix} [\mathbf{1} \quad \mathbf{p}] \right)^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{p}^T \end{bmatrix} = \begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T \mathbf{p} \\ \mathbf{p}^T \mathbf{1} & \mathbf{p}^T \mathbf{p} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{p}^T \end{bmatrix} = \begin{bmatrix} N & 0 \\ 0 & N \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{p}^T \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \\ \frac{1}{N} \mathbf{p}^T \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}}_{ML} = \mathbf{H}^{\#} \mathbf{X} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \\ \frac{1}{N} \mathbf{p}^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \mathbf{X} \\ \frac{1}{N} \mathbf{p}^T \mathbf{X} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \sum_{n=0}^{N-1} X[n] \\ \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n X[n] \end{bmatrix}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

where we exploited the fact that N is even, we have:

$$\mathbf{1}^T \mathbf{1} = \|\mathbf{1}\|^2 = N, \quad \mathbf{p}^T \mathbf{p} = \|\mathbf{p}\|^2 = N, \quad \mathbf{1}^T \mathbf{p} = \mathbf{p}^T \mathbf{1} = \sum_{n=0}^{N-1} (-1)^n = 0.$$

$$\hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \sum_{n=0}^{N-1} X[n], \quad \hat{B}_{ML} = \frac{1}{N} \mathbf{p}^T \mathbf{X} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n X[n]$$

We could obtain the same results by calculating the derivative of $\ln L(A, B)$ with respect to A and B

$$\begin{aligned} \ln L(A, B) &= k - \frac{1}{2\sigma_w^2} (\mathbf{x} - (A\mathbf{1} + B\mathbf{p}))^T (\mathbf{x} - (A\mathbf{1} + B\mathbf{p})) \\ &= k - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{x} + A^2 \mathbf{1}^T \mathbf{1} + B^2 \mathbf{p}^T \mathbf{p} + AB \mathbf{1}^T \mathbf{p} + AB \mathbf{p}^T \mathbf{1} - A \mathbf{1}^T \mathbf{x} - B \mathbf{p}^T \mathbf{x} - A \mathbf{x}^T \mathbf{1} - B \mathbf{x}^T \mathbf{p}) \\ &= k - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{x} + A^2 N + B^2 N - 2A \mathbf{1}^T \mathbf{x} - 2B \mathbf{p}^T \mathbf{x}) \\ &= k_2 - \frac{1}{2\sigma_w^2} (A^2 N + B^2 N - 2A \mathbf{1}^T \mathbf{x} - 2B \mathbf{p}^T \mathbf{x}) \end{aligned}$$

and solving the linear system of 2 equations that we obtain by setting the two derivatives equal to 0.

The estimators are **linear** and so the estimates are **Gaussian** random variables, being $\mathbf{X}$ a Gaussian vector; consequently the estimation errors are Gaussian random variables. To derive their mean values and their variances we can proceed as follows:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \mathbf{X} \\ \frac{1}{N} \mathbf{p}^T \mathbf{X} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T (A\mathbf{1} + B\mathbf{s} + \mathbf{W}) \\ \frac{1}{N} \mathbf{p}^T (A\mathbf{1} + B\mathbf{s} + \mathbf{W}) \end{bmatrix} = \begin{bmatrix} A + \frac{1}{N} \mathbf{1}^T \mathbf{W} \\ B + \frac{1}{N} \mathbf{p}^T \mathbf{W} \end{bmatrix}$$

$$\boldsymbol{\varepsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} A - \hat{A}_{ML} \\ B - \hat{B}_{ML} \end{bmatrix} = \begin{bmatrix} \varepsilon_A \\ \varepsilon_B \end{bmatrix} = \begin{bmatrix} -\frac{1}{N} \mathbf{1}^T \mathbf{W} \\ -\frac{1}{N} \mathbf{p}^T \mathbf{W} \end{bmatrix}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$E\{\boldsymbol{\varepsilon}\} = \begin{bmatrix} E\{\boldsymbol{\varepsilon}_A\} \\ E\{\boldsymbol{\varepsilon}_B\} \end{bmatrix} = \begin{bmatrix} E\left\{-\frac{1}{N}\mathbf{1}^T \mathbf{W}\right\} \\ E\left\{-\frac{2}{N}\mathbf{p}^T \mathbf{W}\right\} \end{bmatrix} = \begin{bmatrix} -\frac{1}{N}\mathbf{1}^T E\{\mathbf{W}\} \\ -\frac{2}{N}\mathbf{p}^T E\{\mathbf{W}\} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Hence, the estimators are **unbiased**.

$$\begin{aligned} MSE\{\hat{\boldsymbol{\theta}}_{ML}\} &= \begin{bmatrix} MSE\{\hat{A}_{ML}\} \\ MSE\{\hat{B}_{ML}\} \end{bmatrix} = \begin{bmatrix} \text{var}\{\hat{A}_{ML}\} \\ \text{var}\{\hat{B}_{ML}\} \end{bmatrix} = \begin{bmatrix} \left(\frac{1}{N}\right)^2 \text{var}\{\mathbf{1}^T \mathbf{X}\} \\ \left(\frac{1}{N}\right)^2 \text{var}\{\mathbf{p}^T \mathbf{X}\} \end{bmatrix} = \begin{bmatrix} \frac{1}{N^2} E\{\mathbf{1}^T \mathbf{W} (\mathbf{1}^T \mathbf{W})^T\} \\ \frac{1}{N^2} E\{\mathbf{p}^T \mathbf{W} (\mathbf{p}^T \mathbf{W})^T\} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{N^2} \mathbf{1}^T E\{\mathbf{W} \mathbf{W}^T\} \mathbf{1} \\ \frac{1}{N^2} \mathbf{p}^T E\{\mathbf{W} \mathbf{W}^T\} \mathbf{p} \end{bmatrix} = \begin{bmatrix} \frac{1}{N^2} \mathbf{1}^T (\sigma_w^2 \mathbf{I}) \mathbf{1} \\ \frac{1}{N^2} \mathbf{p}^T (\sigma_w^2 \mathbf{I}) \mathbf{p} \end{bmatrix} = \begin{bmatrix} \frac{\sigma_w^2}{N} \\ \frac{\sigma_w^2}{N} \end{bmatrix} \end{aligned}$$

$$MSE\{\hat{A}_{ML}\} = MSE\{\hat{B}_{ML}\} = \frac{\sigma_w^2}{N}, \text{ so the estimators are } \mathbf{consistent}.$$

(3) The elements of the Fisher Information Matrix (FIM) are easily obtained by calculating the 2nd order derivative of the log-likelihood function:

$$\ln L(A, B) = k_2 - \frac{1}{2\sigma_w^2} (A^2 N + B^2 N - 2A \mathbf{1}^T \mathbf{x} - 2B \mathbf{p}^T \mathbf{x})$$

$$\frac{d^2 \ln L(A, B)}{dA^2} = -\frac{N}{\sigma_w^2}, \quad \frac{d^2 \ln L(A, B)}{dB^2} = -\frac{N}{\sigma_w^2}, \quad \frac{d^2 \ln L(A, B)}{dA dB} = 0.$$

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma_w^2} & 0 \\ 0 & \frac{N}{\sigma_w^2} \end{bmatrix} \rightarrow CRB(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\sigma_w^2}{N} & 0 \\ 0 & \frac{\sigma_w^2}{N} \end{bmatrix}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$CRB(A) = \frac{\sigma_w^2}{N} = MSE\{\hat{A}_{ML}\}, \quad CRB(B) = \frac{\sigma_w^2}{N} = MSE\{\hat{B}_{ML}\}$$

Hence, the estimators are **efficient**.

(3) The two estimators have the same MSE as the estimator of B ; the reason is that the two signals vectors $\mathbf{1}$ and $\mathbf{p}$ have the same energy.

We require: $\sqrt{E\left\{\left(\frac{B - \hat{B}_{ML}}{B}\right)^2\right\}} < \frac{10}{100} = 10^{-1}$, that implies:

$$E\left\{\left(\frac{B - \hat{B}_{ML}}{B}\right)^2\right\} = \frac{E\left\{(B - \hat{B}_{ML})^2\right\}}{B^2} = \frac{MSE\{\hat{B}_{ML}\}}{B^2} = \frac{\sigma_w^2}{NB^2} < 10^{-2} \quad \rightarrow \quad N > \frac{10^2 \sigma_w^2}{B^2} = \frac{1000}{25} = 40,$$

so we need to collect at least 40 samples.